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I declare that, with the exceptions listed above or directly indicated in my research paper, the work that I have submitted is my own original work.

Signed¹: Khushi Verma

Date: March 5, 2026

¹ Note that a typed name is acceptable in lieu of a signature.

Blip Glitch vs White-Noise Burst Classification with a 1D CNN on Log-FFT Strain

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Abstract

A major challenge for gravitational-wave (GW) searches is contamination of detector strain data by short-duration non-Gaussian transients (glitches), which can increase false alarms. We study a binary classification problem on simulated 1 s single-detector strain segments sampled at 1024 Hz: noise + white-noise burst (WNB) versus noise + blip glitch. Each segment is represented by the log-magnitude of its one-sided Fourier transform and classified using a lightweight one-dimensional convolutional neural network (CNN). The decision threshold is set using negative training examples to target a false positive rate (FPR) of 5%. We evaluate performance across nine WNB signal-to-noise ratio (SNR) bins spanning 5–50 using 1000 examples per class per bin. The measured FPR remains stable across the SNR range (0.030–0.048), while the accuracy stays near chance (0.495–0.511), indicating limited discriminative performance for this model and representation under the evaluated conditions.

I. Introduction

Ground-based laser interferometers such as the Advanced LIGO detectors and Advanced Virgo [3] form a global network for observing gravitational waves (GWs), enabling direct detections and event catalogues [6]. After calibration and data conditioning, these instruments provide a time-series strain channel $h(t)$, which is the primary input for astrophysical searches and follow-up analyses.

For CBC sources, whose waveforms are accurately modelled by general relativity, searches typically employ matched filtering against a bank of template waveforms [8]. In contrast, burst-like transients are often weakly modelled or unmodelled, motivating complementary search strategies that look for excess power in time–frequency representations and use proxy signals for performance studies [4]. A commonly used proxy family is the *white-noise burst* (WNB), a broadband transient, used to represent generic short GW bursts.

A major limitation for both CBC and burst analyses is that detector strain data are not perfectly Gaussian. In addition to stationary noise, the data contain short-duration, non-Gaussian transients known as *glitches* [2]. Glitches can mimic astrophysical burst signals and increase the false-alarm rate, making robust control of false positives operationally important for search pipelines [9]. In this work, we focus on *blip* glitches, a commonly observed glitch class in Advanced LIGO data [5].

Figures 1 and 2 show representative simulated examples used in this study (both at SNR=30). The WNB example ap-

pears as a longer, noise-like transient in the time domain, with broadband support over a chosen frequency band. The blip example is visibly more impulsive: a short-lived spike in $h(t)$, with a comparatively broad frequency-domain footprint that can overlap burst-like power in the detector band. This qualitative similarity in frequency content is exactly why blips are a nuisance for generic burst searches.

We pose a binary classification task on 1 s single-detector strain segments sampled at 1024 Hz: *noise+WNB* (class 0) versus *noise+blip glitch* (class 1). Each segment is represented by the log-magnitude of its one-sided discrete Fourier transform, and we train a lightweight one-dimensional convolutional neural network (CNN) classifier. Performance is summarised using the false positive rate (FPR) and accuracy across WNB signal-to-noise ratio (SNR) bins spanning SNR 5–50, using a fixed decision threshold chosen to control false positives.

II. Data

We construct simulated single-detector strain segments of duration 1 s sampled at 1024 Hz using `MLy` [1]. Each example consists of stationary Gaussian noise with an injected transient waveform, forming two classes:

- **Class 0: noise+WNB**, where the injected waveform is a white-noise burst (WNB) generated and injected using `MLy`.
- **Class 1: noise+blip glitch**, where the injected waveform is a blip glitch generated using `gengli` [7] and injected into noise using the same `MLy` injection pipeline.

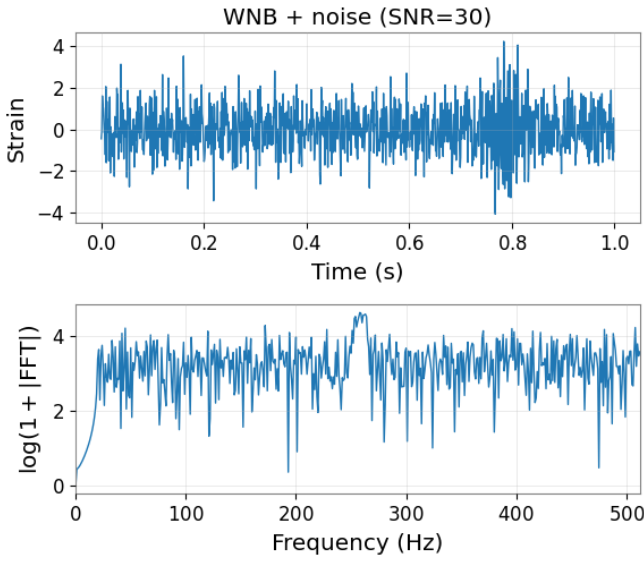


Figure 1: Example class 0 strain segment (noise+WNB) in the time domain (top) and its log-magnitude FFT representation (bottom) at SNR=30.

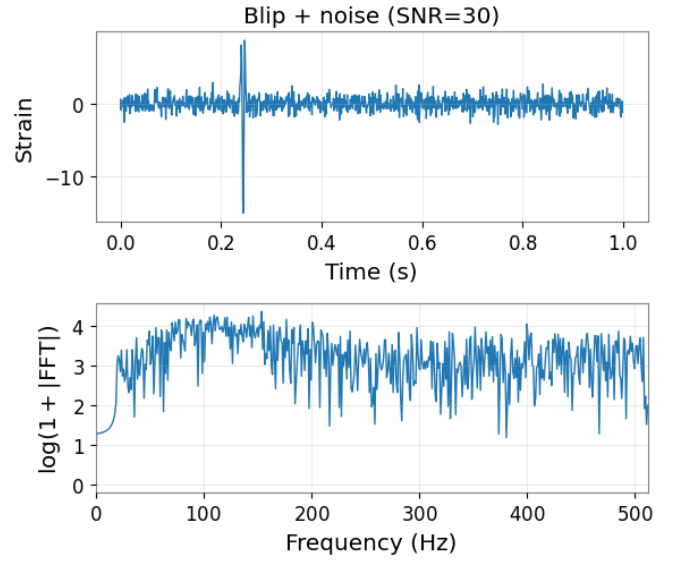


Figure 2: Example class 1 strain segment (noise+blip glitch) in the time domain (top) and its log-magnitude FFT representation (bottom) at SNR=30.

Injection signal-to-noise ratio (SNR) values are drawn uniformly in the range 5–50.

II.1. White-Noise Burst Injections

WNB waveforms are generated with a sample rate of 1024 Hz and randomised time–frequency parameters. The lower and upper frequency bounds are chosen within $f_{\min} = 30$ Hz and $f_{\max} = 400$ Hz, and durations are drawn uniformly between 0.05 s and 0.5 s. These WNB waveforms are injected into 1 s stationary Gaussian-noise segments to form class 0 examples.

II.2. Blip Injections

The generated blip waveforms are injected into 1 s stationary Gaussian-noise segments to form class 1 examples, using the same sampling rate and segment duration as class 0.

II.3. Frequency-domain Representation

For each 1 s strain segment $x(t)$, we standardise the time series to zero mean and unit variance and compute the one-sided discrete Fourier transform. The network input is the log-magnitude spectrum,

$$X(f) = \log(1 + |\text{FFT}\{x(t)\}|), \quad (1)$$

which compresses the range and yields a fixed-length frequency-domain representation.

III. Model and Training

III.1. CNN Architecture

We use a lightweight one-dimensional Convolutional Neural Network (CNN) that takes the log-magnitude frequency-domain representation as input. The network consists of a single 1D convolutional layer with 16 filters (kernel size 7) and ReLU activation, followed by max pooling, global average pooling, and a sigmoid output neuron producing a probability score $p \in [0, 1]$ for class 1 (blip glitch).

III.2. Training Procedure

The training dataset contains 8000 examples in total (4000 per class), generated by injecting WNB waveforms and glitches into stationary Gaussian noise. We split the dataset into training and validation sets using a stratified 80/20 split. The model is trained using the Adam optimiser (learning rate 10^{-3}) with binary cross-entropy loss for 50 epochs and batch size 256.

III.3. Threshold Selection

To control false positives, we choose a fixed decision threshold p_{thr} using negative (class 0) training examples. We set p_{thr} to the 95th percentile of the class 0 training scores, corresponding to a target false positive rate $\text{FPR} = 0.05$ on the training distribution. This threshold is then used for evaluation.

IV. Model Evaluation

We evaluate performance using the false positive rate (FPR) and accuracy across WNB SNR bins spanning SNR 5–50. For

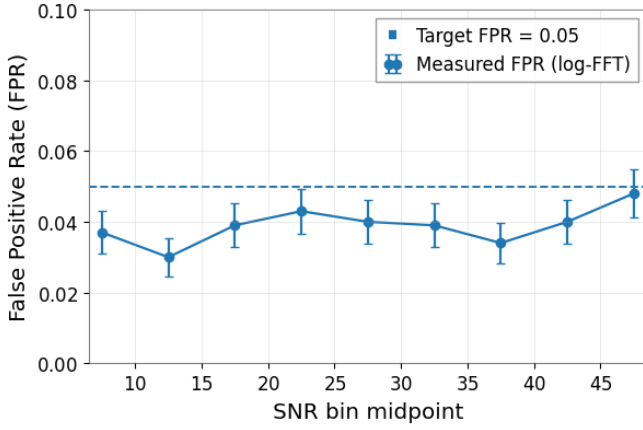


Figure 3: Measured FPR versus SNR bin midpoint for the CNN evaluated on the log-magnitude FFT representation. Error bars show the standard error for $n_0 = 1000$ negative examples per bin. The dashed line indicates the target FPR of 0.05 used to choose p_{thr} from negative training scores.

each SNR bin, we generate 1000 examples per class (class 0: noise+WNB; class 1: noise+blip glitch) by injecting the corresponding waveform into 1 s stationary Gaussian-noise segments and computing the log-magnitude FFT representation. Predicted probabilities are converted to binary labels using the fixed decision threshold p_{thr} (selected from negative training scores), and a confusion matrix is computed in each bin to obtain the FPR and accuracy.

V. Results

Figure 3 shows the measured false positive rate (FPR) as a function of WNB SNR bin midpoint when evaluating the CNN on the log-magnitude FFT representation. The dashed line indicates the target FPR of 0.05 used to select the decision threshold from negative (class 0) training scores. The decision threshold used in evaluation was $p_{\text{thr}} = 0.527$. Across the nine SNR bins spanning 5–50, the measured FPR remains stable and below the target value, ranging from 0.030 to 0.048.

Table 1 summarises the binned evaluation quantitatively, reporting FPR and accuracy for 1000 examples per class per SNR bin. While the FPR is controlled at the few-percent level, the accuracy remains close to chance (~ 0.50) in all bins (0.495–0.511), indicating limited discriminative power for this simple CNN using the FFT-magnitude representation under the evaluated test distribution.

VI. Discussion

The binned evaluation demonstrates that the decision threshold controls false positives effectively: the measured FPR remains stable at the few-percent level across SNR 5–50. However, the accuracy remains close to 0.5 across all bins, suggesting that this simple CNN and the log-magnitude

FFT representation provide limited separation between noise+WNB and noise+blip examples under the evaluated test distribution.

There are several plausible contributors to this behaviour. First, glitches often contain short-lived features confined to a narrow time window, using only the log-magnitude FFT discards phase (timing) information and may reduce sensitivity to these time-localised morphologies; this is consistent with the example strain segments shown in Figs. 1 and 2. Second, the model is intentionally minimal (a single convolutional block), which may be insufficient to learn subtle morphological differences in broadband transients. Finally, this study uses synthetic stationary Gaussian noise in a single detector and focuses on a single glitch family (blips), whereas real detector data contain non-stationary artefacts and a diverse glitch population.

Future work could explore richer time–frequency representations, more expressive architectures, and multi-detector features that exploit coherence. It would also be valuable to broaden the signal and glitch sets (e.g., including cosine-Gaussian bursts, CBC waveforms, and additional glitch classes) to better reflect the diversity of transient noise observed in practice [2].

VII. Conclusion

We investigated a binary classification problem on 1 s single-detector strain segments sampled at 1024 Hz: noise+WNB versus noise+blip glitch. Using a simple 1D CNN trained on the log-magnitude FFT representation and a decision threshold chosen to target an FPR of 0.05 on negative training examples, we evaluated performance across SNR bins spanning 5–50 with 1000 examples per class per bin. The measured FPR remains stable and below 0.05 across all SNR bins, but the accuracy stays near chance (~ 0.50), indicating limited discriminative performance for this model/representation under the evaluated conditions. These results motivate the use of richer time–frequency representations and more expressive model architectures for robust glitch identification in GW data.

Table 1: Binned evaluation results for the FFT-magnitude CNN using a fixed decision threshold $p_{\text{thr}} = 0.527$. For each SNR bin, 1000 examples per class were generated.

SNR bin	FPR	Accuracy
5–10	0.037	0.505
10–15	0.030	0.511
15–20	0.039	0.497
20–25	0.043	0.498
25–30	0.040	0.502
30–35	0.039	0.500
35–40	0.034	0.507
40–45	0.040	0.506
45–50	0.048	0.495

The code for this study is available on GitHub [10].

[10] Khushi Verma. Microproject_CNN_FFT_GW. GitHub repository, 2026. Accessed: 2026-03-05.

VIII. Acknowledgements

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